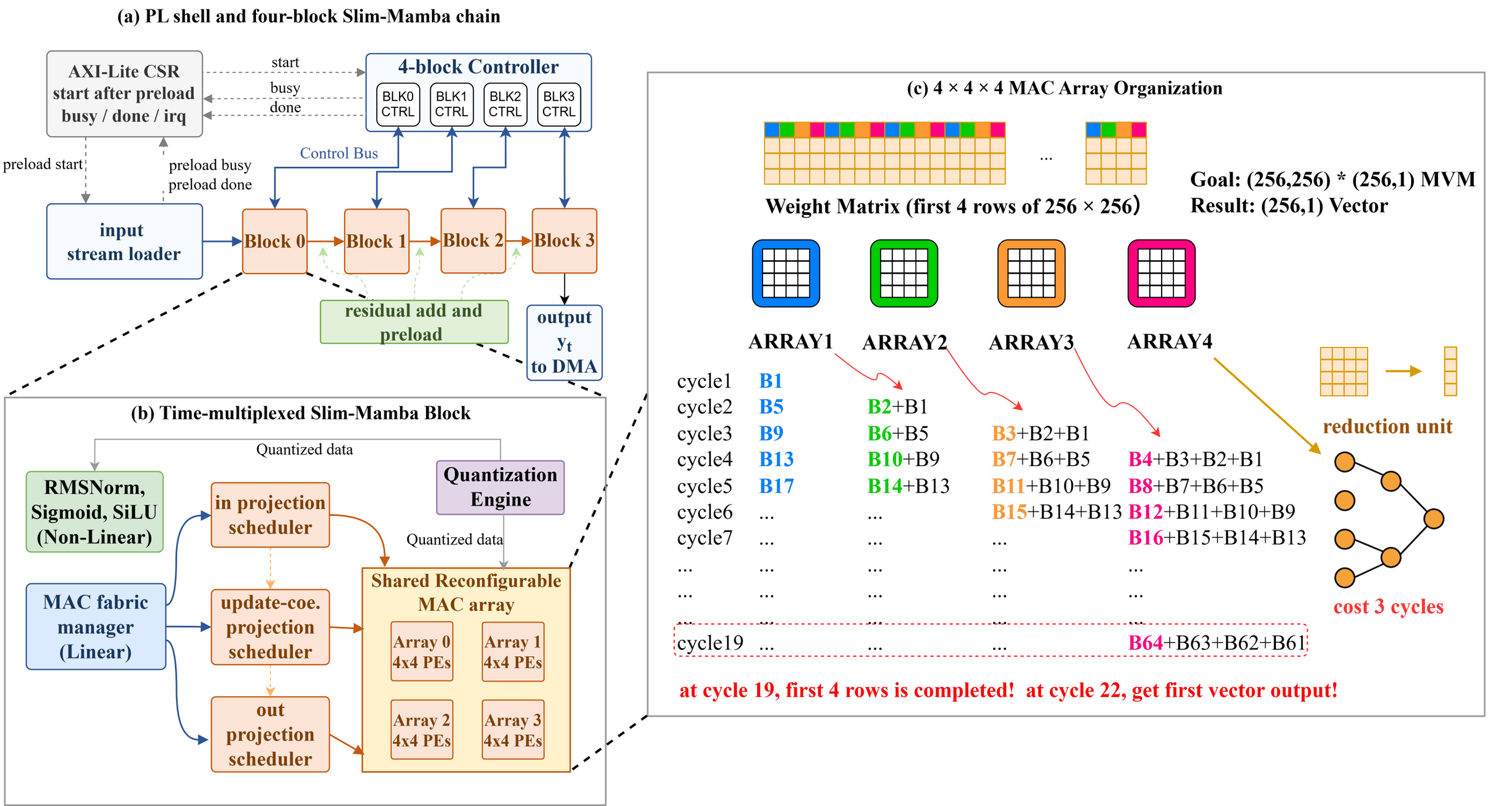

(a) PL shell and four-block Slim-Mamba chain
AXI-Lite CSR
start after preload
busy / done / irq
start
busy
done
4-block Controller
BLK0 CTRL
BLK1 CTRL
BLK2 CTRL
BLK3 CTRL
Control Bus
preload start
preload busy
preload done
input stream loader
Block 0
Block 1
Block 2
Block 3
residual add and preload
output y_t to DMA
(b) Time-multiplexed Slim-Mamba Block
Quantized data
RMSNorm, Sigmoid, SiLU (Non-Linear)
Quantization Engine
in projection scheduler
Quantized data
MAC fabric manager (Linear)
update-coe. projection scheduler
out projection scheduler
Shared Reconfigurable MAC array
Array 0 4x4 PEs
Array 1 4x4 PEs
Array 2 4x4 PEs
Array 3 4x4 PEs
(c) 4 × 4 × 4 MAC Array Organization
Weight Matrix (first 4 rows of 256 × 256)
Goal: (256,256) * (256,1) MVM
Result: (256,1) Vector
ARRAY1
ARRAY2
ARRAY3
ARRAY4
reduction unit
cycle1 B1
cycle2 B5 B2+B1
cycle3 B9 B6+B5 B3+B2+B1
cycle4 B13 B10+B9 B7+B6+B5 B4+B3+B2+B1
cycle5 B17 B14+B13 B11+B10+B9 B8+B7+B6+B5
cycle6 B15+B14+B13 B12+B11+B10+B9
cycle7 B16+B15+B14+B13
cycle19 B64+B63+B62+B61
cost 3 cycles
at cycle 19, first 4 rows is completed! at cycle 22, get first vector output!